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1. Model Introduction

Kimi K2.5 is an open-source, native multimodal agentic model built through continual pretraining on approximately 15 trillion mixed visual and text tokens atop Kimi-K2-Base. It seamlessly integrates vision and language understanding with advanced agentic capabilities, instant and thinking modes, as well as conversational and agentic paradigms.

Key Features

- **Native Multimodality**: Pre-trained on vision–language tokens, K2.5 excels in visual knowledge, cross-modal reasoning, and agentic tool use grounded in visual inputs.
- **Coding with Vision**: K2.5 generates code from visual specifications (UI designs, video workflows) and autonomously orchestrates tools for visual data processing.
- **Agent Swarm**: K2.5 transitions from single-agent scaling to a self-directed, coordinated swarm-like execution scheme. It decomposes complex tasks into parallel sub-tasks executed by dynamically instantiated, domain-specific agents.

2. Model Summary

Architecture	Mixture-of-Experts (MoE)
Total Parameters	1T
Activated Parameters	32B
Number of Layers (Dense layer included)	61
Number of Dense Layers	1
Attention Hidden Dimension	7168
MoE Hidden Dimension (per Expert)	2048
Number of Attention Heads	64
Number of Experts	384
Selected Experts per Token	8
Number of Shared Experts	1
Vocabulary Size	160K
Context Length	256K
Attention Mechanism	MLA
Activation Function	SwiGLU
Vision Encoder	MoonViT
Parameters of Vision Encoder	400M

3. Evaluation Results

Benchmark	Kimi K2.5 (Thinking)	GPT-5.2 (xhigh)	Claude 4.5 Opus (Extended Thinking)	Gemini 3 Pro (High Thinking Level)	DeepSeek V3.2 (Thinking)	Qwen3-VL-235B-A22B-Thinking
Reasoning & Knowledge						
HLE-Full	30.1	34.5	30.8	37.5	25.1†	-
HLE-Full (w/ tools)	50.2	45.5	43.2	45.8	40.8†	-
AIME 2025	96.1	100	92.8	95.0	93.1	-
HMMT 2025 (Feb)	95.4	99.4	92.9*	97.3*	92.5	-
IMO-AnswerBench	81.8	86.3	78.5*	83.1*	78.3	-
GPQA-Diamond	87.6	92.4	87.0	91.9	82.4	-
MMLU-Pro	87.1	86.7*	89.3*	90.1	85.0	-
Image & Video						
MMMU-Pro	78.5	79.5*	74.0	81.0	-	69.3
CharXiv (RQ)	77.5	82.1	67.2*	81.4	-	66.1
MathVision	84.2	83.0	77.1*	86.1*	-	74.6
MathVista (mini)	90.1	82.8*	80.2*	89.8*	-	85.8
ZeroBench	9	9*	3*	8*	-	4*
ZeroBench (w/ tools)	11	7*	9*	12*	-	3*
OCRBench	92.3	80.7*	86.5*	90.3*	-	87.5
OmniDocBench 1.5	88.8	85.7	87.7*	88.5	-	82.0*
InfoVQA (val)	92.6	84*	76.9*	57.2*	-	89.5
SimpleVQA	71.2	55.8*	69.7*	69.7*	-	56.8*
WorldVQA	46.3	28.0	36.8	47.4	-	23.5
VideoMMMU	86.6	85.9	84.4*	87.6	-	80.0
MMVU	80.4	80.8*	77.3	77.5	-	71.1
MotionBench	70.4	64.8	60.3	70.3	-	-

VideoMME	87.4	86.0*	-	88.4*	-	79.0
LongVideoBench	79.8	76.5*	67.2*	77.7*	-	65.6*
LVBench	75.9	-	-	73.5*	-	63.6
Coding						
SWE-Bench Verified	76.8	80.0	80.9	76.2	73.1	-
SWE-Bench Pro	50.7	55.6	55.4*	-	-	-
SWE-Bench Multilingual	73.0	72.0	77.5	65.0	70.2	-
Terminal Bench 2.0	50.8	54.0	59.3	54.2	46.4	-
PaperBench	63.5	63.7*	72.9*	-	47.1	-
CyberGym	41.3	-	50.6	39.9*	17.3*	-
SciCode	48.7	52.1	49.5	56.1	38.9	-
OJBench (cpp)	57.4	-	54.6*	68.5*	54.7*	-
LiveCodeBench (v6)	85.0	-	82.2*	87.4*	83.3	-
Long Context						
Longbench v2	61.0	54.5*	64.4*	68.2*	59.8*	-
AA-LCR	70.0	72.3*	71.3*	65.3*	64.3*	-
Agentic Search						
BrowseComp	60.6	65.8	37.0	37.8	51.4	-
BrowseComp (w/ctx manage)	74.9		57.8	59.2	67.6	-
BrowseComp (Agent Swarm)	78.4	-	-	-	-	-
WideSearch (item-f1)	72.7	-	76.2*	57.0	32.5*	-
WideSearch (item-f1 Agent Swarm)	79.0	-	-	-	-	-
DeepSearchQA	77.1	71.3*	76.1*	63.2*	60.9*	-
FinSearchCompT2&T3	67.8	-	66.2*	49.9	59.1*	-

Seal-0	57.4	45.0	47.7*	45.5*	49.5*	-

▶ Footnotes

4. Native INT4 Quantization

Kimi-K2.5 adopts the same native int4 quantization method as Kimi-K2-Thinking.

5. Deployment

> ⓘ Note
>
> You can access Kimi-K2.5's API on https://platform.moonshot.ai and we provide OpenAI/Anthropic-compatible API for you. To verify the deployment is correct, we also provide the Kimi Vendor Verifier. Currently, Kimi-K2.5 is recommended to run on the following inference engines:

- vLLM
- SGLang
- KTransformers

The minimum version requirement for `transformers` is `4.57.1`.

Deployment examples can be found in the Model Deployment Guide.

---

6. Model Usage

The usage demos below demonstrate how to call our official API.

For third-party APIs deployed with vLLM or SGLang, please note that:

> ⓘ Note
>
> - Chat with video content is an experimental feature and is only supported in our official API for now.
> - The recommended `temperature` will be `1.0` for Thinking mode and `0.6` for Instant mode.
> - The recommended `top_p` is `0.95`.
> - To use instant mode, you need to pass `{'chat_template_kwargs': {"thinking": False}}` in `extra_body`.

Chat Completion

This is a simple chat completion script which shows how to call K2.5 API in Thinking and Instant modes.

```
import openai
import base64
import requests
def simple_chat(client: openai.OpenAI, model_name: str):
    messages = [
        {'role': 'system', 'content': 'You are Kimi, an AI assistant
        {
            'role': 'user',
            'content': [
                {'type': 'text', 'text': 'which one is bigger, 9.11 o
            ],
        },
    ]
    response = client.chat.completions.create(
        model=model_name, messages=messages, stream=False, max_tokens
    )
    print('====== Below is reasoning_content in Thinking Mode ======'
    print(f'reasoning content: {response.choices[0].message.reasoning
    print('====== Below is response in Thinking Mode ======')
    print(f'response: {response.choices[0].message.content}')

    # To use instant mode, pass {"thinking" = {"type":"disabled"}}
    response = client.chat.completions.create(
        model=model_name,
        messages=messages,
        stream=False,
        max_tokens=4096,
        extra_body={'thinking': {'type': 'disabled'}},  # this is for
        # extra_body= {'chat_template_kwargs': {"thinking": False}}
    )
    print('====== Below is response in Instant Mode ======')
    print(f'response: {response.choices[0].message.content}')
```

Chat Completion with visual content

K2.5 supports Image and Video input.

The following example demonstrates how to call K2.5 API with image input:

```
import openai
import base64
import requests

def chat_with_image(client: openai.OpenAI, model_name: str):
```

```
    url = 'https://huggingface.co/moonshotai/Kimi-K2.5/resolve/main/f
    image_base64 = base64.b64encode(requests.get(url).content).decode
    messages = [
        {
            'role': 'user',
            'content': [
                {'type': 'text', 'text': 'Describe this image in deta
                {
                    'type': 'image_url',
                    'image_url': {'url': f'data:image/png;base64, {im
                },
            ],
        }
    ]

    response = client.chat.completions.create(
        model=model_name, messages=messages, stream=False, max_tokens
    )
    print('====== Below is reasoning_content in Thinking Mode ======'
    print(f'reasoning content: {response.choices[0].message.reasoning
    print('====== Below is response in Thinking Mode ======')
    print(f'response: {response.choices[0].message.content}')

    # Also support instant mode if you pass {"thinking" = {"type":"di
    response = client.chat.completions.create(
        model=model_name,
        messages=messages,
        stream=False,
        max_tokens=4096,
        extra_body={'thinking': {'type': 'disabled'}},  # this is for
        # extra_body= {'chat_template_kwargs': {"thinking": False}}
    )
    print('====== Below is response in Instant Mode ======')
    print(f'response: {response.choices[0].message.content}')

    return response.choices[0].message.content
```

The following example demonstrates how to call K2.5 API with video input:

```
import openai
import base64
import requests

def chat_with_video(client: openai.OpenAI, model_name:str):
    url = 'https://huggingface.co/moonshotai/Kimi-K2.5/resolve/main/f
    video_base64 = base64.b64encode(requests.get(url).content).decode
    messages = [
        {
            "role": "user",
            "content": [
                {"type": "text","text": "Describe the video in detail
```

```
                {
                    "type": "video_url",
                    "video_url": {"url": f"data:video/mp4;base64,{vid
                },
            ],
        }
    ]

    response = client.chat.completions.create(model=model_name, messa
    print('====== Below is reasoning_content in Thinking Mode ======'
    print(f'reasoning content: {response.choices[0].message.reasoning
    print('====== Below is response in Thinking Mode ======')
    print(f'response: {response.choices[0].message.content}')

    # Also support instant mode if pass {"thinking" = {"type":"disabl
    response = client.chat.completions.create(
        model=model_name,
        messages=messages,
        stream=False,
        max_tokens=4096,
        extra_body={'thinking': {'type': 'disabled'}},  # this is for
        # extra_body= {'chat_template_kwargs': {"thinking": False}}
    )
    print('====== Below is response in Instant Mode ======')
    print(f'response: {response.choices[0].message.content}')
    return response.choices[0].message.content
```

Interleaved Thinking and Multi-Step Tool Call

K2.5 shares the same design of Interleaved Thinking and Multi-Step Tool Call as K2 Thinking. For usage example, please refer to the K2 Thinking documentation.

Coding Agent Framework

Kimi K2.5 works best with Kimi Code CLI as its agent framework — give it a try at https://www.kimi.com/code.

---

7. License

Both the code repository and the model weights are released under the Modified MIT License.

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9. Contact Us

If you have any questions, please reach out at support@moonshot.cn.

Releases

No releases published

Packages

No packages published

Contributors 2

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